

4.4 Resampling

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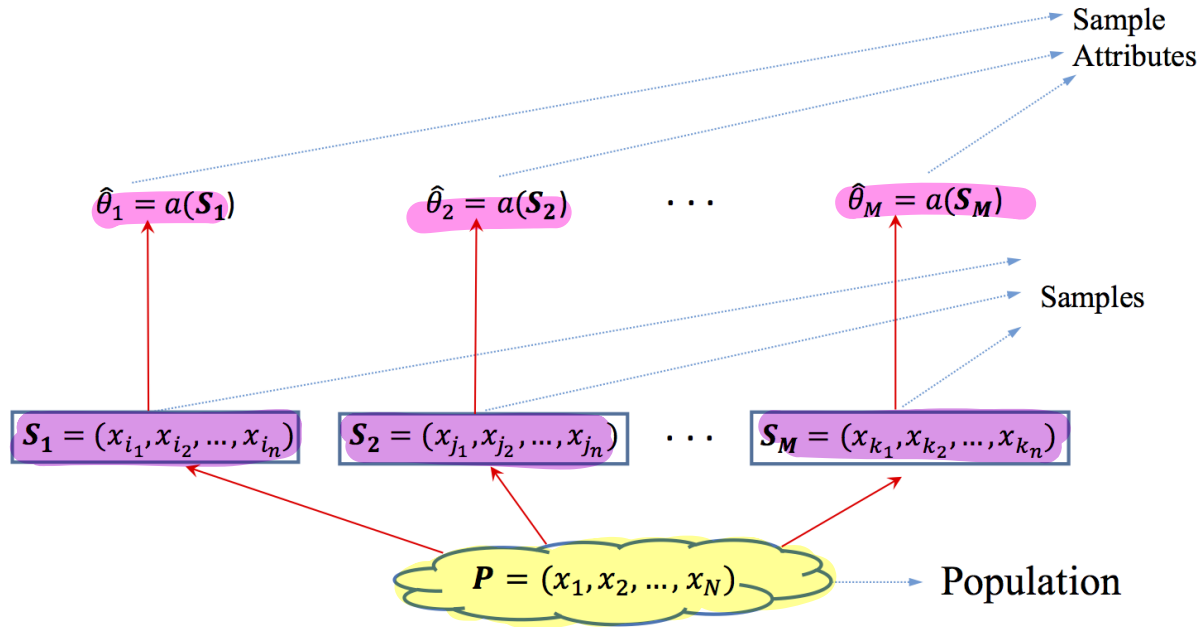
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4.4.0 Resampling Intuition

Overview

- We motivate the idea of resampling and the bootstrap.
-
- As shown in previous sections, understanding the sampling behaviour of sample attributes is essential for making inferences about any population attribute. For example:
 - * Knowing the sampling distribution of a discrepancy measure allows us to test hypotheses
 - * Knowing the sampling distribution of a pivotal quantity allows us to construct confidence intervals
 - To perform inference, we draw a sample $\mathcal{S}$ of size n from a study population $\mathcal{P}$ according to some sampling mechanism
 - then calculate the sample attribute $a(\mathcal{S})$ to estimate its population counterpart $a(\mathcal{P})$.
 - To understand the sampling distribution of the attribute $a(\mathcal{S})$:
 - we draw M samples $\mathcal{S}_1, \dots, \mathcal{S}_M$ and
 - use the values $a(\mathcal{S}_1), \dots, a(\mathcal{S}_M)$ to inform us about the sampling distribution of $a(\mathcal{S})$.

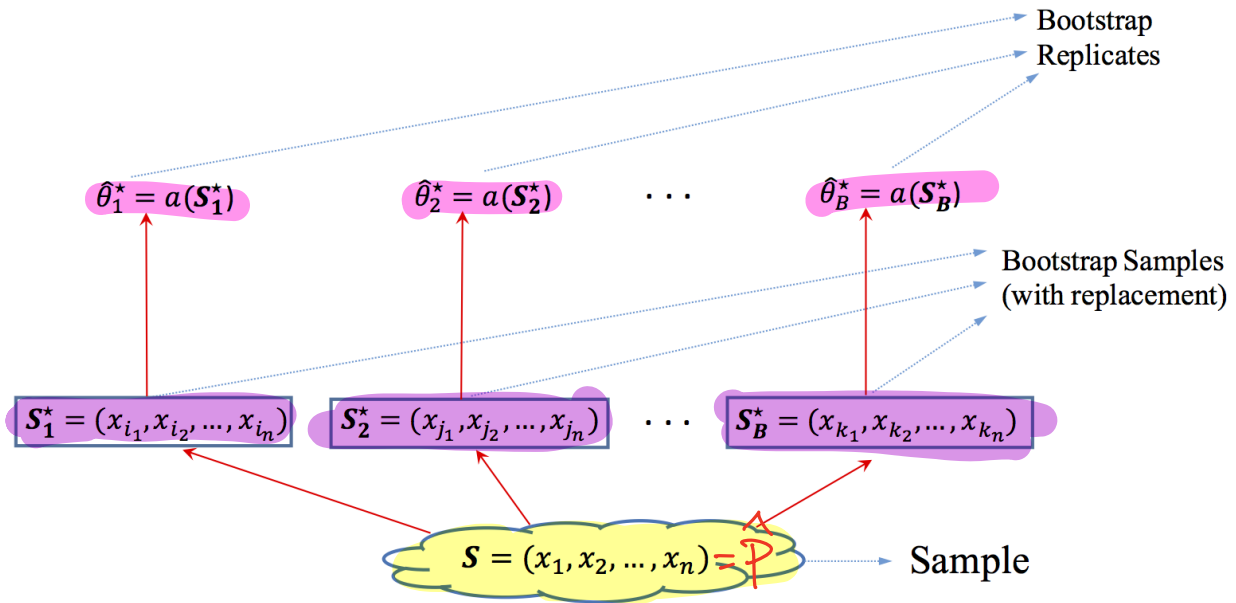


- However, this process requires undertaking repeated sampling from the population
- But in practice, the population from which our sample was taken cannot generally be repeatedly sampled
 - Therefore: we have only one sample.
- **Resampling** methods aim to *mimic* this process by repeatedly sampling S as if it were P
 - In particular, we draw B samples $S_1^*, \dots, S_B^*$ of size n independently from a population $P^* = S$
 - Ideally, P^* would be the study population P , but as already mentioned this would require repeated sampling from the population
 - Instead we use a sample S as an estimate of the population P , i.e., $\hat{P} = S$, or in usual *bootstrap notation* $P^* = S$

- The sample population has only n units, so the without-replacement sampling mechanism will immediately exhaust the population. Therefore, we sample **with replacement**.

$$S_1^*, S_2^*, \dots, S_B^* \neq S \text{ necessarily}$$

- Thus an approximate sampling distribution is obtained by drawing B samples $S_1^*, \dots, S_B^*$ of size n from $\mathcal{P}^*$ with replacement
 - and on each **bootstrap sample** calculate the attribute value: $a(S_1^*), \dots, a(S_B^*)$



Summary

- We saw that bootstrap resampling approximates the sampling distribution of an attribute by using the sample as an estimate of the population, and then repeatedly sampling from it.

4.4.1 The Bootstrap Method

Overview

- Here we see how we can use the bootstrap to estimate quantities of interest from the sampling distribution.

- The distribution of any attribute over the bootstrap samples $\mathcal{S}_i^*$ ($i = 1, 2, \dots, B$) from $\mathcal{P}^*$ is a **bootstrap estimate** of the distribution of the same attribute over all possible samples $\mathcal{S}_i$ from $\mathcal{P}$

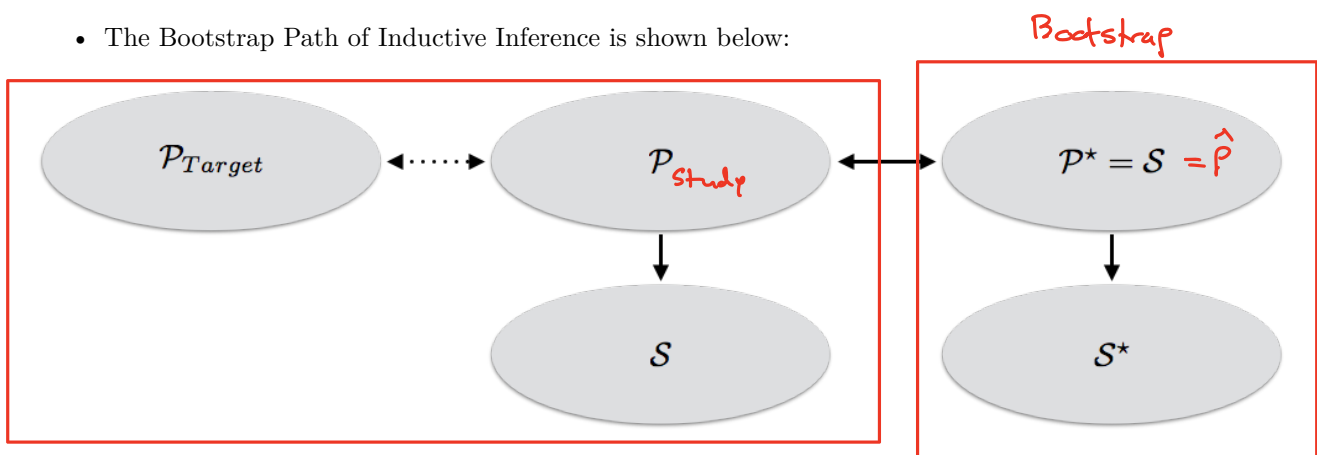
✳ This approach to mimicking the sampling distribution was named the **bootstrap** method when it was first proposed in 1979 by Bradley Efron.

- The word “bootstrap” conveys the notion of starting something up from nothing as in “pulling oneself over a fence by one’s bootstraps”.

– It suggests something for nothing, or something impossible to achieve.

- In Efron (1979), other possible names were *Swiss Army Knife*, *Meat Axe*, *Swan-Dive*, *Jack-Rabbit*.

- The Bootstrap Path of Inductive Inference is shown below:



Why is bootstrapping so important? What does it provide?

With a single sample we're now able to construct an estimate of the sampling distribution for any attribute and not have to make any restrictive (i.e., normal) assumptions.

Shark Length Example: Average

- Throughout this section we will illustrate relevant concepts using the population of shark encounters ($N = 65$) found in the `sharks` data
- The code below constructs an approximate sampling distribution for estimates of the average shark length based on samples of size $n = 6$
 - Note that there are $\binom{N}{n} = \binom{65}{6} = 82,598,880$ possible samples. Our approximate sampling distribution is based on $M = 10,000$ of these

```
sharks <- read.csv("../Data/sharks.csv")
popSharks <- rownames(sharks)

# A function to put n or N in the denominator of SD
# rather than n-1 or N-1
sdn <- function( y.pop ) {
  N = length(y.pop)
  sqrt( var(y.pop)*(N-1)/(N))}

M <- 10^4
n = 6
set.seed(341)
samples <- sapply(1:M, FUN =function(m) sample(popSharks, n, replace = TRUE) )

avePop <- mean(sharks[, "Length"])
avesSamp <- apply(samples, MARGIN = 2,
  FUN = function(s){mean(sharks[s, "Length"])} )
sampleErrors <- avesSamp - avePop

tmpAve <- mean(avesSamp)
tmpSD <- sdn(avesSamp)

sdsSamp <- apply(samples, MARGIN = 2,
  FUN = function(s){sdn(sharks[s, "Length"])} )
```

- With a single sample of $n = 6$ and we are interested in estimating the average shark length.
 - Should we expect the bootstrap to work if n is small?
In general the effectiveness of the bootstrap and the quality of the estimated sampling distribution depends on whether $S = \hat{P}$ is a good approximation of P .
 - The population has only $N = 65$ units so the sample size is a little under 10 percent of the population.
- We draw one sample S drawn from $\mathcal{P}$ using simple random sampling without replacement. The indices of the sampled rows are shown below.

```
getSample <- function(pop, size, replace=FALSE) {
  N <- length(pop)
  pop[sample(1:N, size, replace = replace)]
}
```

```
set.seed(341)
n <- 6
S <- getSample(1:65, n, replace = FALSE)
S
```

```
## [1] 9 20 14 63 13 59
```

$S = \hat{P}$

- Then we draw B bootstrap samples from this single sample.
 - There are $n^n = 6^6 = 46,656$ possible **bootstrap samples** of size $n = 6$ to select.
 - Here, we choose $B = 10,000$ bootstrap samples $S_1^*, \dots, S_{10000}^*$.
 - We might choose a different B depending on the context.

```
Pstar <- S
B <- 10^4
set.seed(341)
Sstar <- sapply(1:B, FUN = function(b) getSample(Pstar, n, replace = T))
```

- We then have a $6 \times B$ matrix of bootstrap sample indices, where each column represents a different sample
 - The first 10 bootstrap samples contain the units shown below

```
dim(Sstar)
```

```
## [1] 6 10000
```

```
Sstar[,1:10]
```

S_1^* S_2^* S_3^* S_{10000}^*

```
##      [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10] . . . . [10000]
## [1,]    9    9   14   20   63   63   13   63   13    63
## [2,]   63   13   14   14   63   13   20    9   13    63
## [3,]   59   20   13   20   63   63    9   20   20    63
## [4,]   59    9    9   20   13   14    9   14   14    59
## [5,]   13   63   59    9    9   20   14   63    9    59
## [6,]   14   14   20   14   63   20   14   14   63    63
```

- We then compute whichever attribute might be of interest on each bootstrap sample.
 - Here we compute the average shark length for each bootstrap sample.

```
avesBootSamp <- sapply(1:B, FUN = function(i) mean(sharks[Sstar[,i], "Length"]))
```

- The initial ten bootstrap sample averages are

```
round(avesBootSamp[1:10], 1)
```

```
## [1] 152.8 138.5 121.7 116.7 170.2 142.3 114.5 124.3 136.3 186.7
```

- The collection of bootstrap averages is an estimate of the sampling distribution of the sample average
 - Note that it could also be thought of as a population of sorts and can thus be summarized like any other population
- Let us compare the **bootstrap estimate** of the sampling distribution to the approximation obtained by repeatedly sampling the population
 - The average shark lengths in the first 10 samples are:

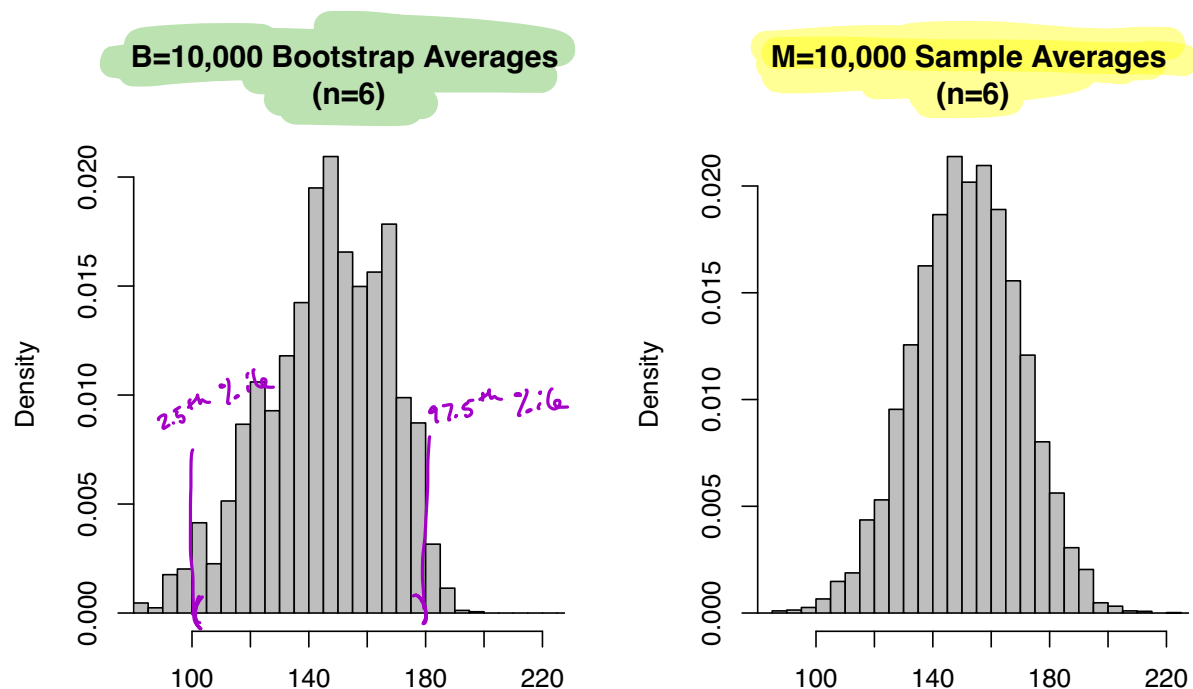
```
round(avesSamp[1:10],1)
```

```
## [1] 132.8 151.5 133.0 177.3 148.8 179.2 170.8 140.5 140.0 169.7
```

- Histograms of both collections of sample estimates are shown below.

```
savePar <- par(mfrow=c(1,2))

hPopAve <- hist(extendrange(c(avesSamp,avesBootSamp)), breaks = 50, plot = FALSE)
hist(avesBootSamp, xlim = range(avesSamp), breaks = hPopAve$breaks,
     freq = FALSE, col = "grey", main = "B=10,000 Bootstrap Averages \n(n=6)", xlab="")
hist(avesSamp, xlim = range(avesSamp), breaks = hPopAve$breaks,
     freq = FALSE, col = "grey", main = "M=10,000 Sample Averages \n(n=6)", xlab="")
```



- Note that the bootstrap estimate is, at best, as good as $a(S)$ (Why?).

If S is weird then $a(S)$ might be very far from $a(P)$.

- As can be seen, the bootstrap distribution gives a sense of how an attribute varies.

✿ Thus, to get an idea of the variability in a sampling distribution we can take the standard deviation of the bootstrap distribution.

↑ helpful for constructing bootstrap CI's

- To easily compare the variability we can construct histograms for each of the following errors:

sample error = $a(S) - a(P)$ ← recentrs the right histogram

bootstrap sample error = $a(S^*) - a(S)$ ← recentrs the left histogram

```
aveSam <- mean(sharks[S, "Length"])

savePar <- par(mfrow=c(1,2))

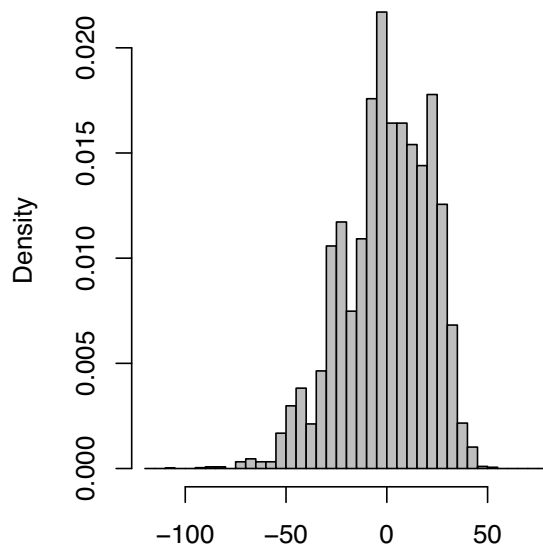
range.avediff <- extendrange(
  c(avesSamp- avePop,
    avesBootSamp- aveSam) )

hPopAvediff <- hist(range.avediff,
  breaks = 50, plot = FALSE)

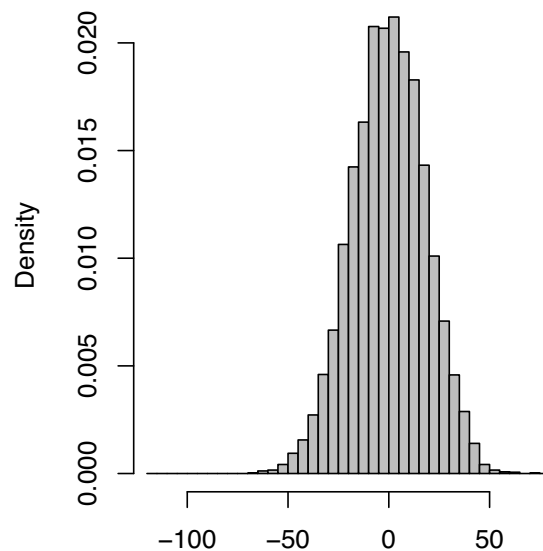
hist(avesBootSamp - aveSam,
  xlim = range.avediff, breaks = hPopAvediff$breaks,
  freq = FALSE, col = "grey",
  main = "B=10,000 Bootstrap Sample Errors \n(n=6)", xlab="")

hist(avesSamp - avePop,
  xlim = range.avediff, breaks = hPopAvediff$breaks,
  freq = FALSE, col = "grey",
  main = "M=10,000 Sample Errors \n(n=6)", xlab="")
```

**B=10,000 Bootstrap Sample Errors
(n=6)**



**M=10,000 Sample Errors
(n=6)**



- The standard deviations in both of these distributions are as follows

```
# Bootstrap Standard Deviation
```

```
sdn(avesBootSamp)
```

```
## [1] 21.32629
```

Estimate of SD of $\bar{Y}$ (i.e., the standard error)

```
# Repeated Sampling Standard Deviation
```

```
sdn(avesSamp)
```

```
## [1] 18.21636
```

Estimate of SD of $\bar{Y}$, but this took 10,000 samples from the population to calculate.

Bootstrap Standard Deviation

- In general, for any attribute $a(\mathcal{P})$, the standard deviation of the corresponding sample attribute's estimator can be estimated from the bootstrap distribution with

$$\hat{SD}_*[\tilde{a}(S)] = \sqrt{\frac{\sum_{b=1}^B [a(S_b^*) - \bar{a}^*]^2}{B-1}}$$

where $\bar{a}^* = \frac{1}{B} \sum_{b=1}^B a(S_b^*)$ is the average attribute value overall B bootstrap samples.

- ✂ Since B is usually large, it does not make any practical difference whether we use B or $B - 1$ in the denominator of the standard deviation.

- This is an estimate of the standard deviation of the sampling distribution for the attribute $a(\mathcal{S})$, which we called the **standard error**.

Special Case: Inference for a Population Average

- In the special case of the arithmetic average $a(\mathcal{S}) = \frac{1}{n} \sum_{u \in \mathcal{S}} y_u$ the bootstrap estimate of its standard deviation is

$$\hat{SD}_*[\bar{y}] = \sqrt{\frac{\sum_{b=1}^B (\bar{y}_b^* - \bar{y}^*)^2}{B-1}}$$

where $\bar{y}^* = \frac{1}{B} \sum_{b=1}^B \bar{y}_b^*$ and $\bar{y}_b^*$ is the average variate value in bootstrap sample $b=1, 2, \dots, B$.

- But we also know that the standard deviation can be estimated with

$$\hat{SD}[\bar{y}] = \frac{\hat{\sigma}}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}} \quad \text{where} \quad \hat{\sigma} = \sqrt{\frac{\sum_{u \in \mathcal{S}} (y_u - \bar{y})^2}{n}}$$

- Calculating these two estimates in the context of the **shark** example yields:

```
n=6
N=dim(sharks)[1]
c(sdn(avesBootSamp), sdn(sharks[S, "Length"])/sqrt(n)*sqrt((N-n)/(N-1)))
```

```
## [1] 21.32629 20.50388
```

- Note:** if the sample is not a good representation of the population, these two numbers might be quite different.

- To more fully compare the two methods of estimating standard deviation we randomly select $M = 2000$ samples of size $n = 6$ from the population, and for each sample:
 - calculate $\frac{\hat{\sigma}}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}$, yielding 2000 repeated-sample-based estimates of $\widehat{SD}[\bar{Y}]$,
 - and generate $B = 200$ bootstrap samples used to obtain 2000 bootstrap estimates of $\widehat{SD}[\bar{Y}]$.

```
set.seed(341)
numSamps <- 2000
n <- 6

sampSes <- sapply(1:numSamps, FUN = function(i) getSample(popSharks, n))
B <- 200

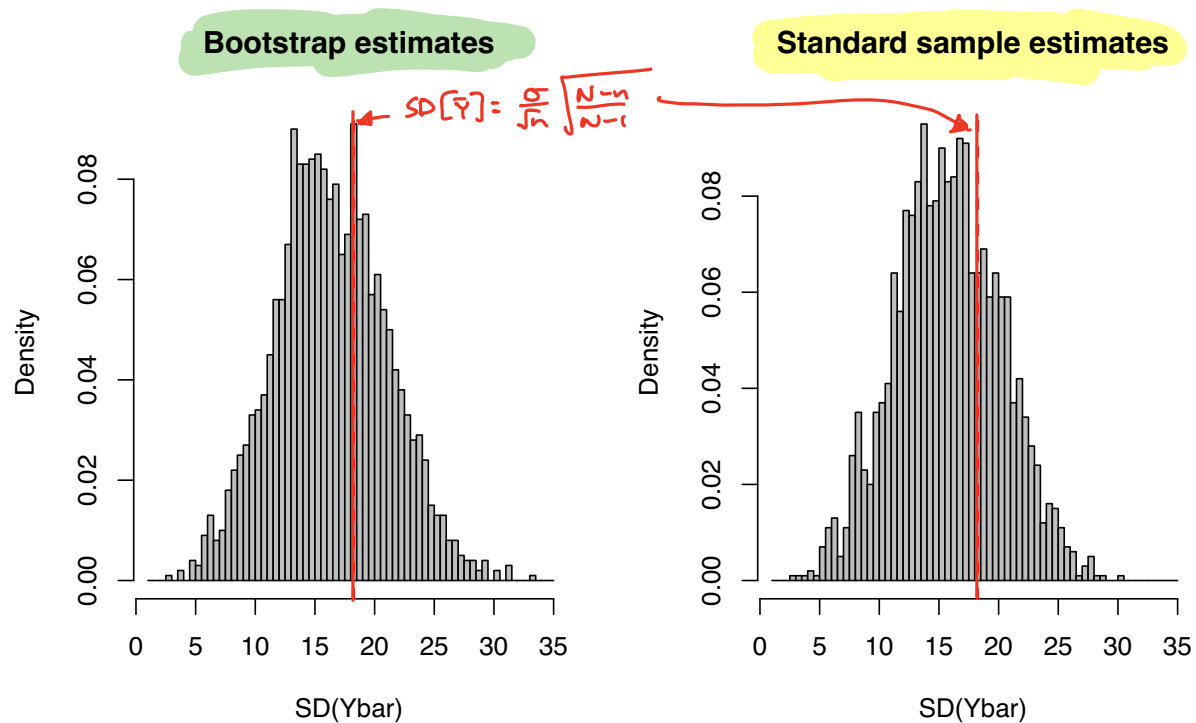
valsBoot = t(apply(sampSes, 2, function(sam=NULL, B=NULL) {
  avesTemp = unlist(mapply( function(i) {
    mean(sharks[sample(sam, n, replace = TRUE), "Length"]) }, 1:B) )

  c( sdn(avesTemp), sdn( sharks[sam, "Length"] )/sqrt(n)*sqrt((N-n)/(N-1)))
}, B=B ))

colnames(valsBoot) <- c("sdBoot", "sdStandard")

par(mfrow=c(1,2))
hTmp <- hist(extendrange(valsBoot), breaks = 50, plot = FALSE)
hist(valsBoot[, "sdBoot"], xlim = extendrange(valsBoot),
      breaks = hTmp$breaks, freq = FALSE, col = "grey",
      main = "Bootstrap estimates", xlab="SD(Ybar)")
abline(v=sdn(avesSamp), col="red", lty=2, lwd=2)

hist(valsBoot[, "sdStandard"], xlim = extendrange(valsBoot),
      breaks = hTmp$breaks, freq = FALSE, col = "grey",
      main = "Standard sample estimates", xlab="SD(Ybar)")
abline(v=sdn(avesSamp), col="red", lty=2, lwd=2)
```

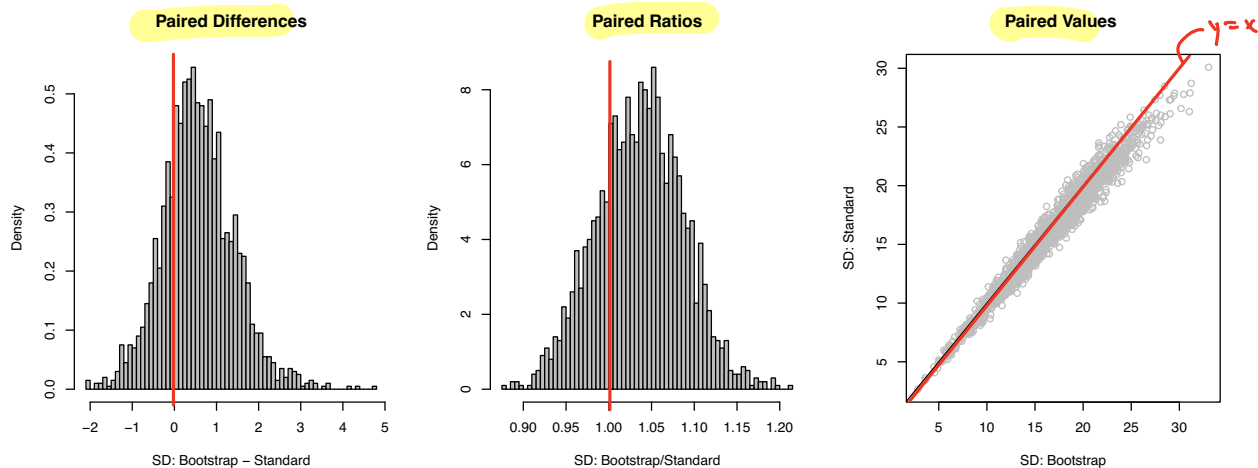


- The two histograms are not too dissimilar in shape, but the bootstrap estimator of standard deviation has produced, on average, slightly larger values than the standard approach.
- This can be more easily seen by constructing plots of the pairs of 2000 standard deviation estimates:

```
par(mfrow=c(1,3))
hist(valsBoot[, "sdBoot"] - valsBoot[, "sdStandard"],
     main = "Paired Differences", breaks=50, col="grey",
     freq=FALSE, xlab="SD: Bootstrap - Standard")

hist(valsBoot[, "sdBoot"] / valsBoot[, "sdStandard"],
     main = "Paired Ratios", breaks=50, col="grey",
     freq=FALSE, xlab="SD: Bootstrap/Standard")

plot(valsBoot[, "sdBoot"], valsBoot[, "sdStandard"],
     main = "Paired Values ", col="grey", xlab="SD: Bootstrap",
     ylab="SD: Standard")
abline(a=0, b=1)
```



A Comment on n Versus $n - 1$

- In the calculations above the function `sdn(...)` was introduced and used which is an implementation of

$$\hat{\sigma} = \sqrt{\frac{\sum_{u \in \mathcal{S}} (y_u - \bar{y})^2}{n}}$$

which has n as a divisor. The built-in function `sd(...)` in R is an implementation of

$$\hat{\sigma} = \sqrt{\frac{\sum_{u \in \mathcal{S}} (y_u - \bar{y})^2}{n - 1}}$$

- * For bootstrap confidence interval calculations, a divisor of n is preferred (since we are treating the sample as a population)
 - This version has the advantage of being **replication invariant**.
 - Replication invariant estimates are preferred and are often called **plug in estimates** in the bootstrap literature e.g. see Efron and Tibshirani (1994) .
- However, when n is reasonably large, there will be little practical difference between the two.

Shark Length Example: Median, Standard Deviation and Skewness

- In the previous investigation we saw that the bootstrap approach to calculating standard deviations of sampling distributions closely matched the (computationally less intensive) approach that exploited some theory specific to the estimation of population averages.
- The huge advantage that the bootstrap approach has is that it is applicable to **any** sample attribute not just the arithmetic average.

- In this section we also consider the

- median
 - standard deviation
 - skewness

- Using the sample (9, 20, 14, 63, 13, 59) and $B = 10,000$ bootstrap samples we can construct bootstrap estimates of each attribute's sampling distribution and hence standard deviation

- The sample estimates of median, standard deviation, and skewness are respectively:

```
skew <- function(z) { 3*(mean(z) - median(z))/sdn(z) }
sam.len = sharks[S, "Length"]
c(median(sam.len), sdn(sam.len), skew(sam.len))
```

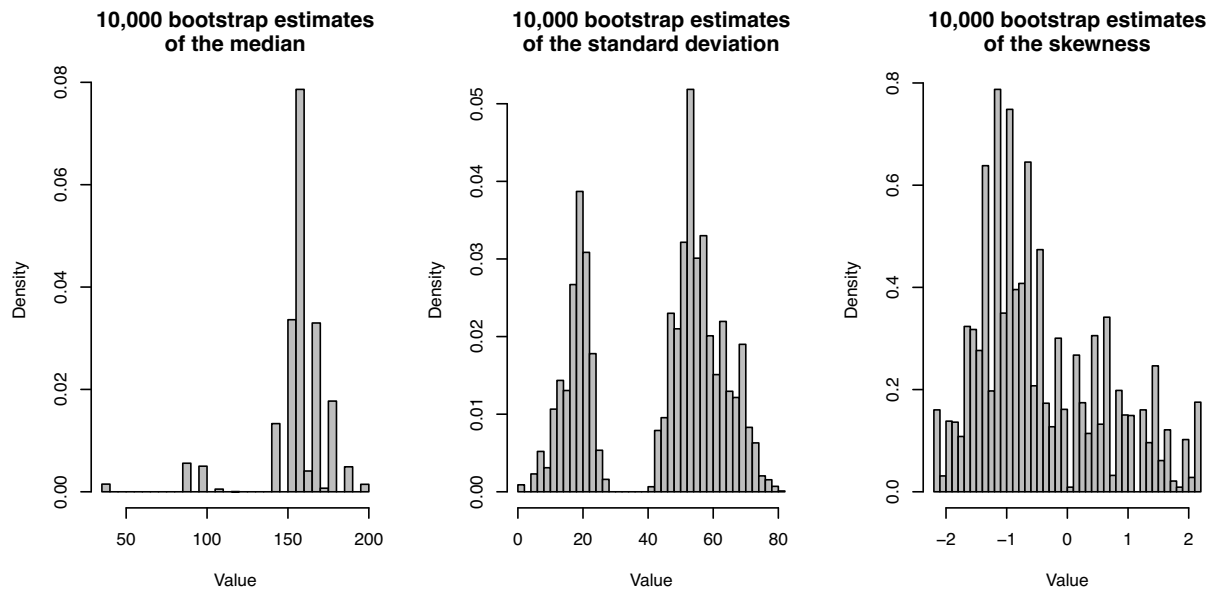
```
## [1] 157.0000000 52.3089115 -0.6499849
```

- Generating $B = 10,000$ bootstrap samples and calculating each attribute on every sample yields the following estimated sample distributions:

```
bootAttr <- apply(Sstar, 2, function(sam){
  pop = sharks[sam, "Length"]
  c(median(pop), sdn(pop), skew(pop))})
par(mfrow=c(1,3))
namHist = c("10,000 bootstrap estimates\nof the median",
            "10,000 bootstrap estimates\nof the standard deviation",
            "10,000 bootstrap estimates\nof the skewness")

for(i in 1:3){hist(bootAttr[i, ], main = namHist[i], xlab="Value",
                  breaks=50, col="grey", freq=FALSE)}
```

3x10000 matrix
of median, SD, and
skewness values on
all B=10,000 bootstrap samples



- The corresponding bootstrap estimate of the standard deviation for each sampling distribution is:

```
apply(X = bootAttr, MARGIN = 1, FUN = sd, na.rm = TRUE)
```

```
## [1] 21.412114 19.799716 1.023634
```

↑ bootstrap standard errors for each of these three attributes.

So What are Bootstrap Distributions Good For?

Estimating (and Correcting for) Sampling Bias

- Recall that the **sampling bias** of an attribute is defined as:

$$\text{Sampling Bias} = E[a(\mathcal{S})] - a(\mathcal{P})$$

- We can use the bootstrap to estimate sampling bias via

$$\widehat{\text{Sampling Bias}} = \text{average bootstrap sample error} = \bar{a}^* - a(\mathcal{S})$$

where $\bar{a}^* = \frac{1}{B} \sum_{b=1}^B a(\mathcal{S}_b^*)$ is the average of the attribute over the bootstrap samples $\mathcal{S}_1^*, \dots, \mathcal{S}_B^*$.

- When an estimator is biased we would like to “correct” it
 - i.e. make a biased estimator unbiased.

- If, in theory, $a(S)$ was biased and we knew the bias then we could subtract the correction from our attribute to make a new attribute $a^*(S)$ that is unbiased:

$$\tilde{a}^*(S) = \tilde{a}(S) - \text{sampling bias}$$

- To verify that $\tilde{a}^*(S)$ that is unbiased:

$$\begin{aligned} E[\tilde{a}^*(S)] &= E[\tilde{a}(S) - \text{sampling bias}] \\ &= E[\tilde{a}(S)] - E[E[\tilde{a}(S)] - a(P)] \\ &= E[\tilde{a}(S)] - E[\tilde{a}(S)] + a(P) = a(P) \end{aligned}$$

- However, we don't typically *know* the sampling bias, so instead we could use the bootstrap estimate of it.

- The bias-corrected bootstrap estimate is

$$\begin{aligned} a^*(S) &= a(S) - \text{sampling bias} \\ &= a(S) - [\bar{a}^* - a(S)] \\ &= 2a(S) - \bar{a}^* \quad \leftarrow \text{approximately unbiased} \end{aligned}$$

Confidence Intervals & Hypothesis Tests

- As was demonstrated in previous sections of the notes, the sampling distribution of an attribute was integral in calculating confidence intervals and performing hypothesis tests.
- Approximate confidence intervals and hypothesis tests can now also be based on the bootstrap estimate of a sampling distribution.
- We explore bootstrap-based confidence in the next section of the notes.
- Bootstrap-based hypothesis tests are not covered in STAT 341.

Summary

- We can use the bootstrap to estimate the standard error of the sampling distribution for *any* attribute.
- We can also estimate the sampling bias.

4.4.2a Bootstrap Confidence Intervals

Overview

- We see how to use the bootstrap to calculate a confidence interval when the bootstrap distribution is approximately normal.
-

- The bootstrap distribution provides a proxy for the sampling distribution for any sample attribute $a(\mathcal{S})$.

- So we can use it to construct (at least approximate) confidence intervals for the unknown population attribute $a(\mathcal{P})$.

↑ coverage of a $100(1-p)\%$ CI might not be exactly $1-p$.

Naive Normal-Theory Intervals

- Recall that confidence intervals for sample averages tend to have the following structure:

$$\left(\bar{y} - c \times \hat{SD}[\bar{Y}], \bar{y} + c \times \hat{SD}[\bar{Y}] \right)$$

- Under an assumption of normality we might pick c such that $P(Z \leq c) = 1-p/2$ generates a $100(1-p)\%$ confidence interval

- If the bootstrap distribution is approximately normal, rather than estimating $\widehat{SD}(\bar{Y})$ by $\frac{\hat{\sigma}}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}$
 - we might estimate $\widehat{SD}(\bar{Y})$ using the standard deviation of the bootstrap distribution of $\bar{Y}$.

- The attraction of this approach, if it works, is that the same approach could be used for any attribute $a(S)$.

- A 95% **normal bootstrap interval** for a population attribute $a(P)$ is

$$\rightarrow a(S) \pm 1.96 * \hat{SD}_* [\hat{a}(S)]$$

- **Question:** What does 95% mean here?

95% of intervals constructed in this way should contain the true value $a(P)$.

Example: Average Shark Length

- A 95% normal bootstrap confidence interval for the average shark length is:

```
sharks <- read.csv("../Data/sharks.csv")
popSharks <- rownames(sharks)

avePop <- mean(sharks[, "Length"])

set.seed(341)
n <- 6
S <- getSample(1:65, n, replace = FALSE)
S
```

```
## [1] 9 20 14 63 13 59
```

```
Pstar <- S
B <- 10^4
set.seed(341)
Sstar <- sapply(1:B, FUN = function(b) getSample(Pstar, n, replace = T))
avesBootSamp <- sapply(1:B, FUN = function(i) mean(sharks[Sstar[,i], "Length"]))
```

```
mean(sharks[S, "Length"]) + 1.96*c(-1,1)*sdn(avesBootSamp)
```

$\nwarrow a(S)$
 $\nwarrow \hat{SD}_*[\hat{a}(S)]$

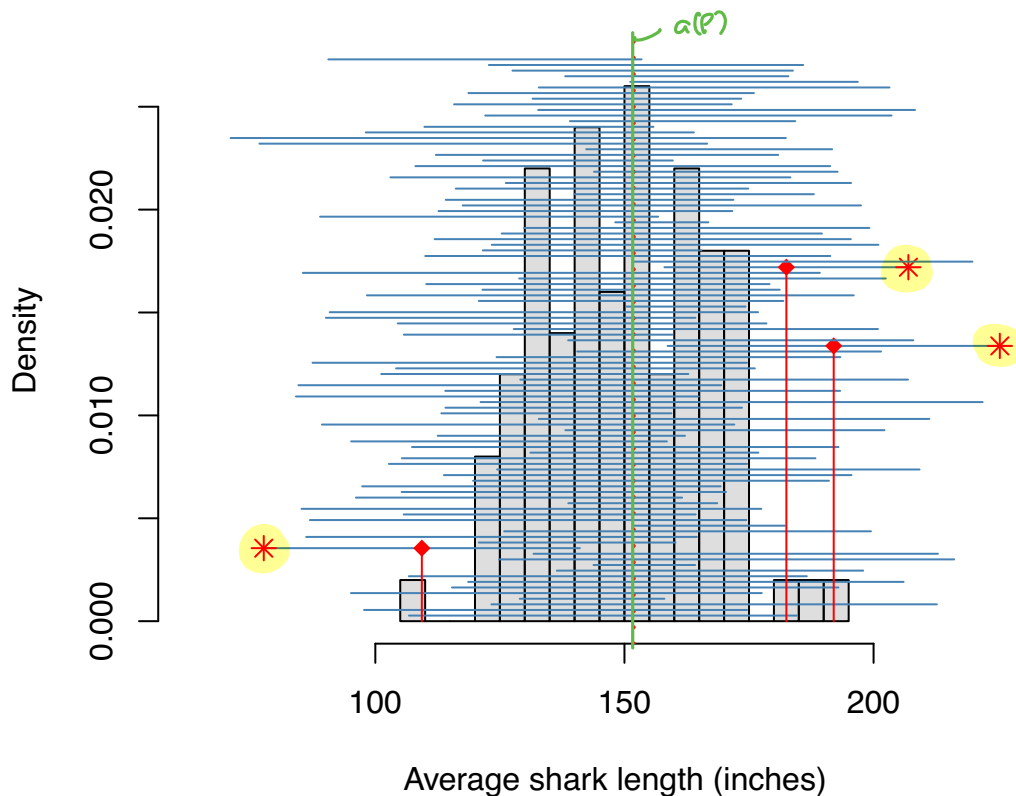
```
## [1] (103.8671, 187.4662)
```

- Note, the population average is 151.86 = $a(P)$

Estimating Coverage Probability: The Normal Bootstrap Interval

- If we use the normal bootstrap interval then we should investigate its properties such as determining if we obtain the proper coverage probability.
- The average shark length of great whites was 151.86 inches (in our population of $N = 65$ encounters).
- Suppose we are interested in a 95% confidence interval for the population mean.
- To get an estimate of the coverage probability for the bootstrap confidence interval, we generate 100 samples of size $n = 6$ and for each sample:
 - generate $B = 1000$ bootstrap samples to obtain the bootstrap estimate of the standard deviation, $\widehat{SD}(\bar{Y})$
 - and then construct the 95% confidence interval (using $c = 1.96$)
- These 100 intervals (and their coverage) are illustrated below

100 individual 95% bootstrap confidence intervals



- As can be seen, 97 of the 100 bootstrap intervals actually cover the average in the population.

Summary

- We saw if the bootstrap distribution is approximately normal then we can construct a confidence interval that exploits this normality.

Exercises

From the Inference Exercises try the standard bootstrap interval part of problem 5.

4.4.2b Bootstrap- t Confidence Intervals

This content is covered in Tutorial 10. See those notes.

Overview

- We use the bootstrap to approximate the sampling distribution of a t -like pivotal quantity.
 - Then we use this approximation to construct a confidence interval. It turns out to be similar to using the t -distribution to approximate the sampling distribution of a pivotal quantity.
-

Introduction

- When $a(\mathcal{S}) = \bar{y}$, we have seen that the quantity

$$Z = \frac{\tilde{a}(\mathcal{S}) - a(\mathcal{P})}{\widehat{SD}[\tilde{a}(\mathcal{S})]}$$

- is approximately pivotal and
- its sampling distribution (over all possible samples) is well approximated by a t -density.

- We can confirm this approximation, using simulation by
 - generating $\mathcal{S}_1, \dots, \mathcal{S}_M$
 - then for each sample calculate

$$Z_i = \frac{a(\mathcal{S}_i) - a(\mathcal{P})}{\widehat{SE}[a(\mathcal{S}_i)]} = \frac{a(\mathcal{S}_i) - a(\mathcal{P})}{\widehat{SD}[\tilde{a}(\mathcal{S}_i)]}$$

- Then an estimate of t_{n-1} can be constructed with $\{Z_1, \dots, Z_M\}$.

- This means we require an estimate of the standard deviation of the estimator, i.e., a standard error.
- e.g. when $a(\mathcal{S})$ is the average, one estimate is

$$SE[\tilde{a}(\mathcal{S}_i)] = \widehat{SD}[\tilde{a}(\mathcal{S}_i)] = \frac{\hat{\sigma}}{\sqrt{n}} \times \sqrt{\frac{N-n}{N-1}}$$

- For example, when $a(\mathcal{P})$ is average shark length, the sampling distribution of the pivotal quantity with $n = 6$ is shown below.
- Overlaid on this histogram is the $t_{(5)}$ density.

```
sharks <- read.csv("../Data/sharks.csv")
popSharks <- rownames(sharks)

# A function to put n or N in the denominator of SD
# rather than n-1 or N-1
sdn <- function( y.pop ) {
  N = length(y.pop)
  sqrt( var(y.pop)*(N-1)/(N) ) }

se.avg <- function(y.sam) {
  sdn(y.sam)/sqrt(length(y.sam)) * sqrt((65-length(y.sam))/(65-1) )
}

popSize <- function(pop) {
  if (is.vector(pop))
  {if (is.logical(pop))
    ## then assume TRUE values identify units
    sum(pop) else length(pop)}
  else nrow(pop)
}

getSample <- function(pop, size, replace=FALSE) {
  N <- popSize(pop)
  pop[sample(1:N, size, replace = replace)]
}

M <- 10^4
n = 6
set.seed(341)
samples <- sapply(1:M, FUN =function(m) sample(popSharks, n, replace = FALSE) )

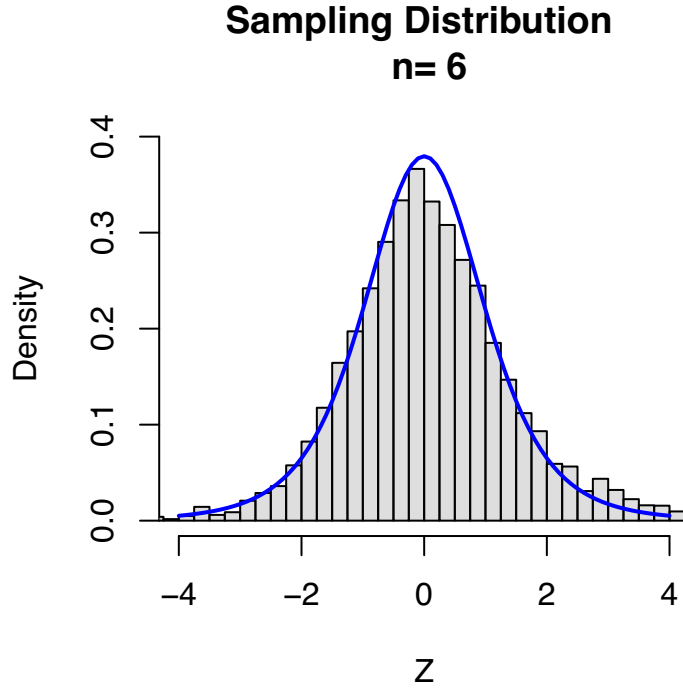
avePop <- mean(sharks[, "Length"])
avesSamp <- apply(samples, MARGIN = 2,
  FUN = function(s){mean(sharks[s, "Length"])})

SEaveSamp <- apply(samples, MARGIN = 2,
  FUN = function(s){se.avg(sharks[s, "Length"])})

ZPop <- (avesSamp - mean(sharks[, "Length"]))/SEaveSamp
```

```
brk = seq(-50,50, by=0.25)
hist(ZPop, freq = FALSE, breaks = brk, col = adjustcolor("grey", 0.5),
     main = paste("Sampling Distribution \n n=", n),
     xlab = "Z", xlim=c(-4,4), ylim=c(0, .4) )

lines(x = seq(-4,4,0.1), y = dt(x = seq(-4,4,0.1), df = n-1), col = "blue", lwd = 2)
```



- Because this follows a t_{n-1} distribution, for a $p \in (0, 1)$ we can find a t_{lower} and t_{upper} such that

$$\begin{aligned} 1 - p &= Pr \left(t_{lower} \leq \frac{\tilde{a}(\mathcal{S}) - a(\mathcal{P})}{\widehat{SD}[\tilde{a}(\mathcal{S})]} \leq t_{upper} \right) \\ &= Pr \left(\tilde{a}(\mathcal{S}) - t_{upper} \times \widehat{SD}[\tilde{a}(\mathcal{S})] \leq a(\mathcal{P}) \leq \tilde{a}(\mathcal{S}) - t_{lower} \times \widehat{SD}[\tilde{a}(\mathcal{S})] \right) \end{aligned}$$

- This suggests that a confidence interval can be constructed with

$$\left[\tilde{a}(\mathcal{S}) - t_{upper} \times \widehat{SD}[\tilde{a}(\mathcal{S})], \tilde{a}(\mathcal{S}) - t_{lower} \times \widehat{SD}[\tilde{a}(\mathcal{S})] \right]$$

- But since the t-distribution is symmetric we have that

$$t_{upper} = -1 \times t_{lower} = c$$

– then the confidence interval becomes

$$\tilde{a}(\mathcal{S}) \pm c \times \widehat{SD}[\tilde{a}(\mathcal{S})]$$

- The t -distribution approximation of a sampling distribution may work for certain attributes, but not all of them.
 - This requires $\tilde{a}(\mathcal{S})$ to be approximately normal over all possible samples. (When might this be true?)
 - If $a(\mathcal{P})$ is the median or a measure of skewness, we would not expect the t -distribution to be a good approximation.

- Instead of approximating the sampling distribution with a t -distribution, we suppose that

$$Z = \frac{\tilde{a}(\mathcal{S}) - a(\text{mathcal{P}})}{\widehat{SD}[\tilde{a}(\mathcal{S})]} \sim F$$

- Then a confidence interval can be constructed with

$$\left[a(\mathcal{S}) - f_{upper} \times \widehat{SD}[\tilde{a}(\mathcal{S})], a(\mathcal{S}) - f_{lower} \times \widehat{SD}[\tilde{a}(\mathcal{S})] \right]$$

- We can estimate F by simulation by

- generating $\mathcal{S}_1, \dots, \mathcal{S}_M$
- then for each sample calculate

$$Z_i = \frac{a(\mathcal{S}_i) - a(\mathcal{P})}{\widehat{SD}[\tilde{a}(\mathcal{S}_i)]}$$

- Then an estimate of F can be constructed with $\{Z_1, \dots, Z_M\}$
- Again this means we require an estimate of the standard deviation of the estimator, i.e., the standard error.

- We have seen that the bootstrap may be used to approximate the sampling distribution of $\tilde{a}(\mathcal{S})$.

- Here we will use the bootstrap to estimate the sampling distribution of

$$Z = \frac{\tilde{a}(\mathcal{S}) - a(\mathcal{P})}{\widehat{SD}[\tilde{a}(\mathcal{S})]}$$

and use it to construct confidence intervals.

- On the **negative** side: this requires computation.
- On the **positive** side: the bootstrap automatically adjusts its shape (and hence quantiles, etc.) to the form of $\tilde{a}(\mathcal{S})$.

- To use the bootstrap to approximate the sampling distribution of Z , we estimate
 - the population $\mathcal{P}$ with the estimate $\mathcal{P}^*$ (the sample $\mathcal{S}$)
 - the sample $\mathcal{S}$ with the bootstrap sample $\mathcal{S}^*$
 - then we generate $\mathcal{S}_1^*, \dots, \mathcal{S}_B^*$ and calculate

$$Z_b^* = \frac{\tilde{a}(\mathcal{S}_b^*) - a(\mathcal{S})}{\widehat{SD}[\tilde{a}(\mathcal{S}_b^*)]}.$$

- The bootstrap estimate of the sampling distribution is $\{Z_1^*, \dots, Z_B^*\}$.
- Then using a $p \in (0, 1)$ and the bootstrap estimate of the sampling distribution we can find Z_{lower}^* and Z_{upper}^* such that

$$1 - p = Pr(Z_{lower}^* \leq Z^* \leq Z_{upper}^*) \approx Pr(Z_{lower}^* \leq Z \leq Z_{upper}^*)$$

- Then a confidence interval using the bootstrap estimate of the pivotal quantity is

$$\left[a(\mathcal{S}) - Z_{upper}^* \times \widehat{SD}[\tilde{a}(\mathcal{S})], a(\mathcal{S}) - Z_{lower}^* \times \widehat{SD}[\tilde{a}(\mathcal{S})] \right]$$

- **Note:** Z_{lower}^* and Z_{upper}^* are quantiles from $\{Z_1^*, \dots, Z_B^*\}$

- Let's compare the sampling distribution to the bootstrap estimate of the sampling distribution for the average (for a given sample).

```
n = 6
M <- 10^4

samSharks <- sample(popSharks, n, replace = FALSE)

samStar <- sapply(1:M, FUN = function(m) sample(samSharks, n, replace = TRUE) )

aveSam <- mean(sharks[samSharks, "Length"])

avesBoot <- apply(samStar, MARGIN = 2,
                  FUN = function(s){mean(sharks[s, "Length"])})

SEaveBoot <- apply(samStar, MARGIN = 2,
                  FUN = function(s){se.avg(sharks[s, "Length"]) })

ZBoot <- (avesBoot - aveSam)/SEaveBoot
```

- The plots below compare these two approaches:

```

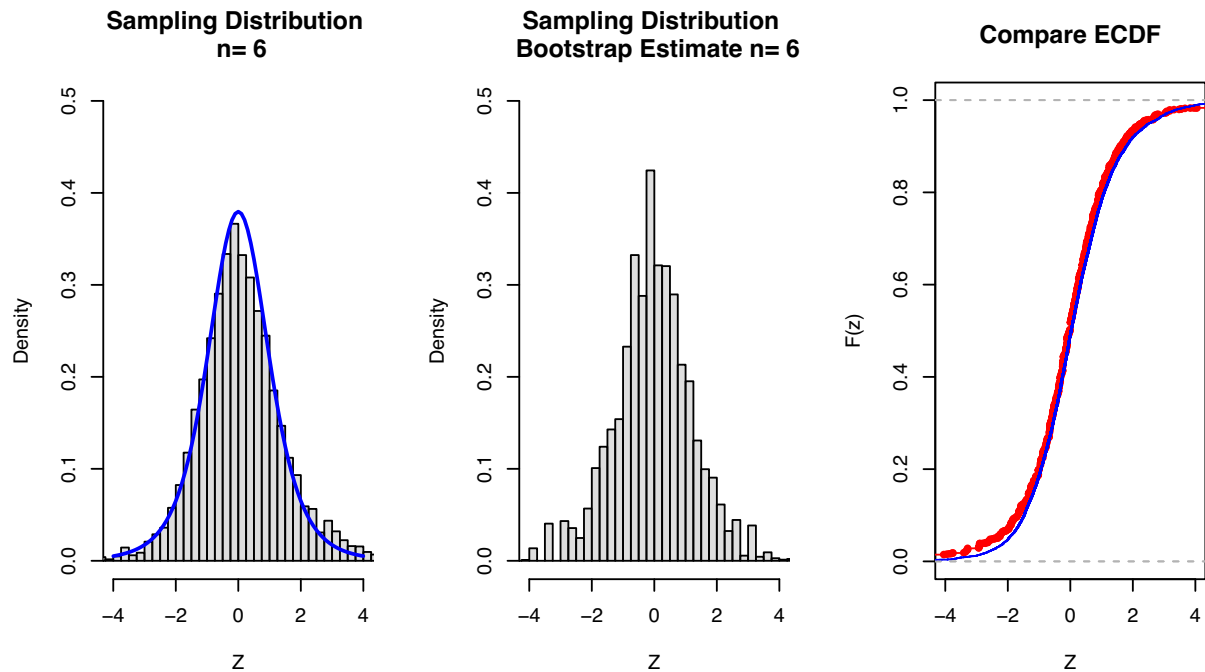
savePar <- par(mfrow=c(1,3))

brk = seq(-60,60, by=0.25)
hist(ZPop, freq = FALSE, breaks = brk, col = adjustcolor("grey", 0.5),
     main = paste("Sampling Distribution \n n=", n),
     xlab = "Z", xlim=c(-4,4), ylim=c(0, .5) )
lines(x = seq(-4,4,0.1), y = dt(x = seq(-4,4,0.1), df = n-1), col = "blue", lwd = 2)

hist(ZBoot, freq = FALSE, breaks = brk, col = adjustcolor("grey", 0.5),
     main = paste("Sampling Distribution \n Bootstrap Estimate n=", n),
     xlab = "Z", xlim=c(-4,4), ylim=c(0, .5) )

plot( ecdf(ZBoot), xlim=c(-4,4), col = "red",
     main = "Compare ECDF", xlab="Z", ylab = "F(z)" )
lines( ecdf(ZPop), col = "blue")

```



- The values of c obtained by each method are:
 - t -distribution:
 - * $t_{upper} = Q_z(0.975) = 2.57$
 - * $t_{lower} = Q_z(0.025) = -2.57$
 - Estimating the sampling distribution directly (many samples):
 - * $c_{upper} = Q_z(0.975) = 3.23$

- * $c_{lower} = Q_z(0.025) = -2.49$
- Bootstrap estimate of the sampling distribution (one sample):
 - * $Z_{upper}^* = Q_z(0.975) = 3.08$
 - * $Z_{lower}^* = Q_z(0.025) = -3.29$

The General Approach

- For a given sample $\mathcal{S}$, attribute $a(\mathcal{S})$, and standard error $\widehat{SD}[\tilde{a}(\mathcal{S})]$
 - Calculate $a(\mathcal{S})$ and $\widehat{SD}[\tilde{a}(\mathcal{S})]$
 - Generate B bootstrap samples $\mathcal{S}_1^*, \dots, \mathcal{S}_B^*$ from $\mathcal{S}$
- For each of the B bootstrap samples from above:
 - Calculate $a(\mathcal{S}_b^*)$ and $\widehat{SD}[\tilde{a}(\mathcal{S}_b^*)]$ and then

$$z_b^* = \frac{a(\mathcal{S}_b^*) - a(\mathcal{S})}{\widehat{SD}[\tilde{a}(\mathcal{S}_b^*)]}$$

- From the values $z_1^*, \dots, z_B^*$, find
 - $c_{lower} = Q_z(p/2)$
 - and $c_{upper} = Q_z(1 - p/2)$
- Then a $100(1 - p)\%$ bootstrap- t confidence interval is

$$\left[a(\mathcal{S}) - c_{upper} \times \widehat{SD}[\tilde{a}(\mathcal{S})], a(\mathcal{S}) - c_{lower} \times \widehat{SD}[\tilde{a}(\mathcal{S})] \right]$$

- **Note** the signs and positions of the constants c_{lower} and c_{upper} in the interval definition.

- **A Note** on standard errors;
 - This method (so far) requires an analytic form to calculate $\widehat{SD}[\tilde{a}(\mathcal{S})]$ (i.e, the standard deviation of the estimator given a single sample)
 - Another interval can be constructed using the bootstrap estimate of the standard error

$$\widehat{SD}_\star[\tilde{a}(\mathcal{S})]$$

- When the bootstrap is used to calculate

$$\widehat{SD}_\star[\tilde{a}(\mathcal{S}_b^\star)]$$

this called the **double** bootstrap which we will look at in Section 4.4.2c.

Example: Average Shark Length

- A Bootstrap- t confidence interval using $\widehat{SD}[\tilde{a}(\mathcal{S})]$ for the average shark length is

```
samSharksLengths = sharks[samSharks, "Length"]

zStar.lower = quantile(ZBoot, 0.025 )
zStar.upper = quantile(ZBoot, 0.975 )

as.numeric(round(mean(samSharksLengths) - c(zStar.upper, zStar.lower)*se.avg(samSharksLengths),2))

## [1] 87.54 194.70
```

- A Bootstrap- t confidence interval using $\widehat{SD}_\star[\tilde{a}(\mathcal{S})]$ for the average shark length is

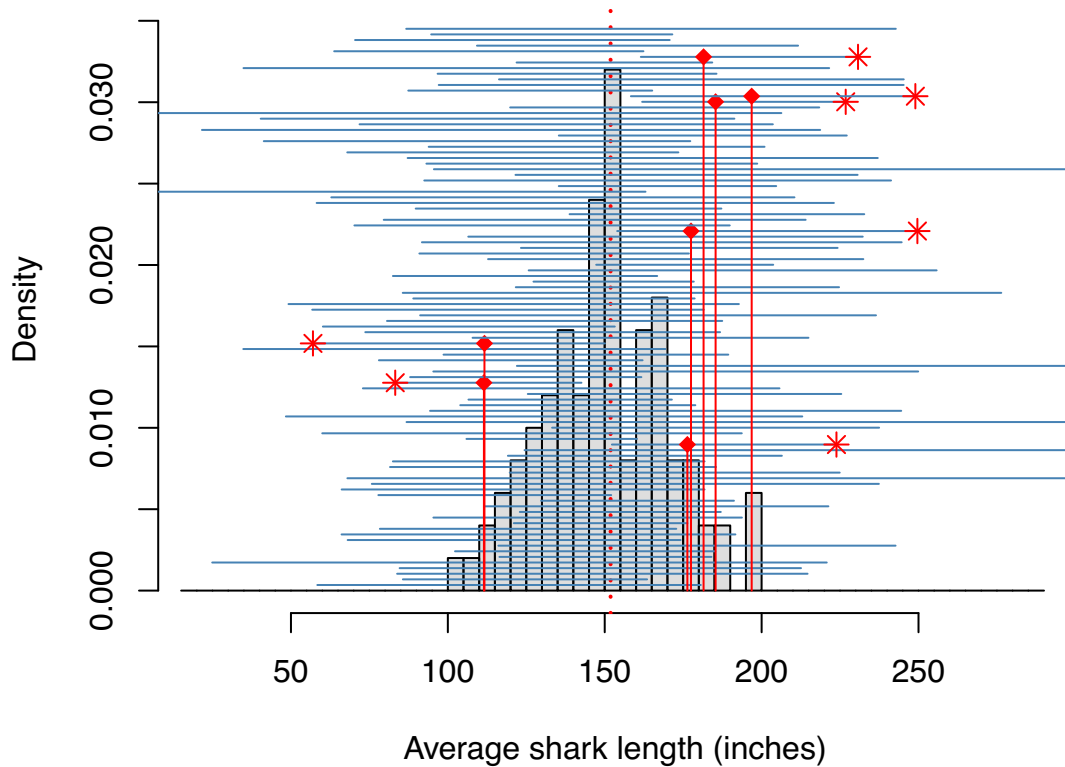
```
as.numeric(round(mean(samSharksLengths) - quantile(ZBoot, c(0.975, 0.025) )*sd(avesBoot),2))

## [1] 85.32 197.07
```

Estimating Coverage Probability: Bootstrap- t Intervals

- Here we investigate the coverage of confidence intervals calculated in this way

100 individual 95% bootstrap-t confidence intervals



- We see that 93 of the 100 bootstrap intervals cover the average in the population.

Summary

- We saw that we can use the bootstrap distribution to approximate the sampling distribution of a pivotal quantity, and we saw that we can construct a confidence interval using it.
- The resulting procedure is similar to using the t-distribution to approximate the sampling distribution of a pivotal quantity.
- But we require an estimate of the standard error of the attribute.

4.4.2c The Double Bootstrap

This content is covered in Tutorial 10. See those notes.

Overview

- We saw that we can use the bootstrap distribution to approximate the sampling distribution of a pivotal quantity, and we saw that we can construct a confidence interval using it.
- But we require an estimate of the standard error of the attribute.
- Here we explore a procedure in which we use the bootstrap to obtain an estimate of the standard error.

The Double Bootstrap

- When $a(\mathcal{S}) = \bar{y}$ we have an analytic form for its standard deviation:

$$SD[\bar{Y}] = \frac{\sigma}{\sqrt{n}} \times \sqrt{\frac{N-n}{N-1}}$$

where replacing σ by $\hat{\sigma}$ gives an estimate $\widehat{SD}[\bar{Y}]$ based on the sample values y_u for $u \in \mathcal{S}$.

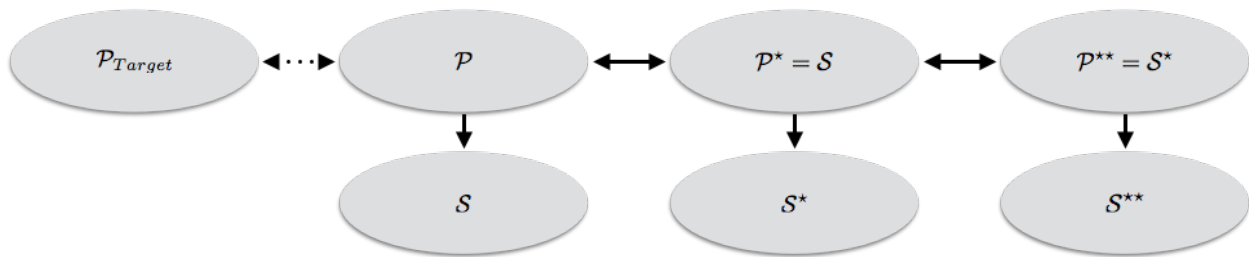
- More generally, when $a(\mathcal{S})$ is a Horvitz-Thompson estimate we also have an analytic form for $\widehat{SD}[\tilde{a}(\mathcal{S})]$
- However, very often an analytic solution is not available for $\widehat{SD}[\tilde{a}(\mathcal{S})]$
 - In which case an estimate can be obtained by using the bootstrap by generating $\mathcal{S}_1^*, \dots, \mathcal{S}_B^*$ and calculating

$$\widehat{SD}_*[\tilde{a}(\mathcal{S})] = \sqrt{\frac{\sum_{b=1}^B (a(\mathcal{S}_b^*) - \bar{a}^*)^2}{B-1}}$$

where $\bar{a}^* = \frac{1}{B} \sum_{b=1}^B a(\mathcal{S}_b^*)$

- But in the bootstrap- t , we need an estimate of $SD[\tilde{a}(\mathcal{S}_b^*)]$ for each bootstrap sample $\mathcal{S}_b^*$!
- In order to determine $\widehat{SD}[\tilde{a}(\mathcal{S}_b^*)]$ for each bootstrap sample $\mathcal{S}_b^*$ we use the **double bootstrap**
 - We apply the bootstrap method to each bootstrap sample $\mathcal{S}_b^*$.

- To apply a **bootstrap within a bootstrap** for each bootstrap sample $\mathcal{S}_b^*$
 - We generate D bootstrap samples, $\mathcal{S}_1^{**}, \dots, \mathcal{S}_D^{**}$, each with replacement from a population now defined as $\mathcal{P}^{**} = \mathcal{S}_b^*$.
 - The standard deviation of the corresponding values $a(\mathcal{S}_1^{**}), \dots, a(\mathcal{S}_D^{**})$ will provide the estimate $\widehat{SD}_*[a(\mathcal{S}_b^*)]$
- This estimate $\widehat{SD}_*[a(\mathcal{S}_b^*)]$ is then substituted into **The General Approach** for bootstrap- t confidence intervals
- Just as with the original bootstrap, this additional bootstrap adds another link in the chain of the inductive path as shown below.



Example: Average Shark Length

- The `bootstrap_t_interval` function defined below is a general-purpose function that may be used to calculate a bootstrap- t confidence interval for *any* attribute.
 - The function requires as input an attribute function `a` which will calculate $a(S)$ for any $S \subset \mathcal{P}$.
 - In the case of the average shark length, this can be written as

```
a <- function(S) {mean(sharks[S, "Length"])}

```

- Here is the `bootstrap_t_interval` function:

```

bootstrap_t_interval <- function(S, a,
                                confidence = 0.95,
                                B = 1000, D = 30){
  ## Here S is the sample, a is a scalar-valued function a(S) of a sample S
  ## which returns the value for S of that attribute of interest
  ## confidence is the level of confidence
  ## B is the outer bootstrap count of replicates used to
  ## calculate the lower and upper limits
  ## D the inner bootstrap count of replicates used to
  ## estimate the standard deviation of the sample attribute
  ## for each (outer) bootstrap sample
  Pstar <- S
  aPstar <- a(Pstar)
  ## get (outer) bootstrap values
  bVals <- sapply(
    1:B,
    FUN = function(b) {
      Sstar <- getSample(Pstar, sampleSize, replace = TRUE)
      aSstar <- a(Sstar)
      ## get (inner) bootstrap values to
      ## estimate the SD
      Pstarstar <- Sstar
      SD_aSstar <- sdn(
        sapply(1:D,
              FUN = function(d){
                Sstarstar <- getSample(Pstarstar, sampleSize, replace = TRUE)
                ## return the attribute value
                a(Sstarstar)
              }
            )
      )
      z <- (aSstar - aPstar)/SD_aSstar
      ## Return the two values
      c(aSstar = aSstar, z = z)
    })
  SDhat <- sdn(bVals["aSstar",])
  zVals <- bVals["z",]
  ## Now use these zVals to get the lower and upper
  ## c values.
  cValues <- quantile(zVals,
                      probs = c((1 - confidence)/2, (confidence + 1)/2),
                      na.rm = TRUE)
  cLower <- min(cValues)
  cUpper <- max(cValues)
  interval <- c(lower = aPstar - cUpper * SDhat,
                middle = aPstar,
                upper = aPstar - cLower * SDhat)
  interval
}

```


- A 95% bootstrap- t confidence interval for the average shark length is

```
sharks <- read.csv("../Data/sharks.csv")
popSharks <- rownames(sharks)
avePop <- mean(sharks[, "Length"])

sampleSize <- 6

S <- getSample(popSharks, size = 6, replace = TRUE)
bootstrap_t_interval(S, a)
```

```
##      lower      middle      upper
## 141.6402 158.0000 209.6687
```

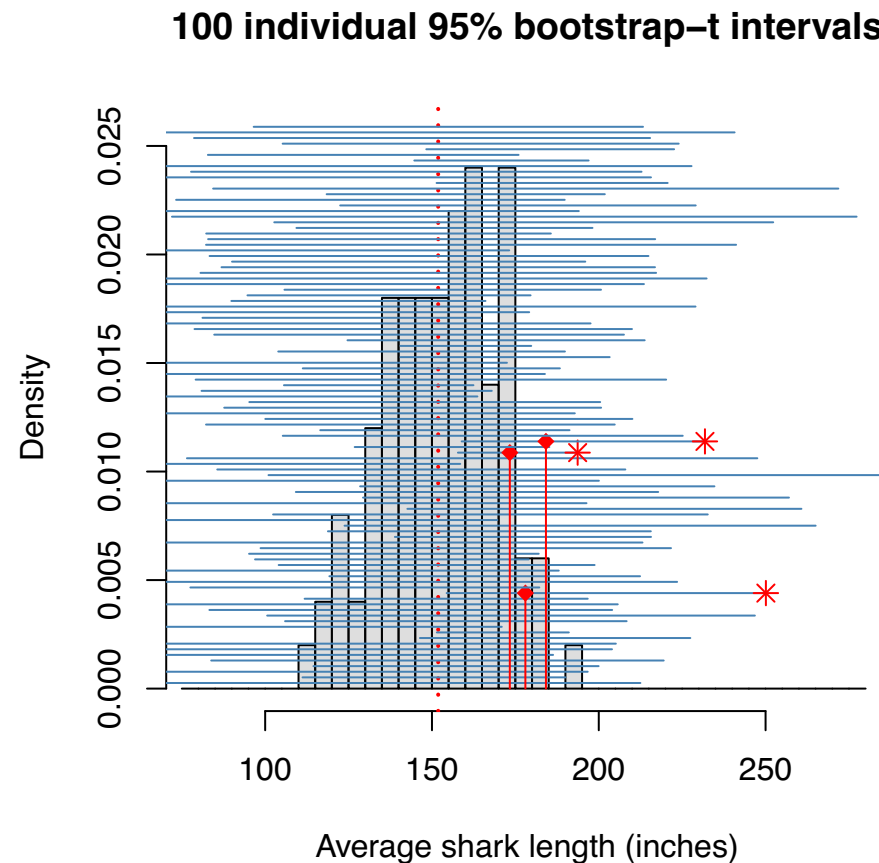
Estimating Coverage Probability: The Double Bootstrap

- Here we investigate the coverage of confidence intervals calculated in this way
 - Note that this simulation takes a decent amount of time to complete

```
numIntervals <- 100
samps <- sapply(1:numIntervals,
               FUN = function(i) getSample(1:65, sampleSize))

confidence <- 0.95
B <- 1000
set.seed(341)

intervals <- sapply(1:numIntervals,
                   FUN = function (i) {
     bootstrap_t_interval(samps[,i], a = a,
                          confidence = confidence, B=B, D=30) } )
```



- We see that 97 of the 100 bootstrap intervals cover the population attribute.

Comments:

- According to [Efron and Tibshirani (1994), pp. 161-162] the bootstrap- t intervals are best suited for attributes which measure “location parameters” like the average, the median, a particular quantile, etc.
- For attributes that do not measure location, it may be necessary to first transform the attribute to a scale that produces bootstrap- t intervals that have good coverage probabilities,
 - Or consider using the percentile method.

Summary

- The double bootstrap can be used to construct confidence intervals by approximating the sampling distribution of a pivotal quantity where the bootstrap is again used to estimate the standard error for each bootstrap sample.

4.4.2d - The Percentile Method

Overview

- We introduce the percentile method for bootstrap-based confidence intervals.

- The sampling distribution of $\tilde{a}(\mathcal{S})$ can be estimated using a sample $\mathcal{S}$ and the bootstrap.

* So why not simply use quantiles from the bootstrap distribution to directly construct a confidence interval?

- The **percentile method** for bootstrap confidence intervals is the following:

- For a given sample $\mathcal{S}$ generate B bootstrap samples $S_1^*, S_2^*, \dots, S_B^*$ by sampling with replacement from $\mathcal{S}$.
- For the b^{th} bootstrap sample ($b = 1, \dots, B$), calculate $a_b = a(S_b^*)$
- From the values $a_1, \dots, a_B$, find $a_{lower} = Q_a(P/2)$ and $a_{upper} = Q_a(1 - P/2)$
- Then the $100(1 - p)\%$ confidence interval is $[a_{lower}, a_{upper}]$

Ex: We want to calculate a CI for $SD(P)$ given a CI for $Var(P)$.

* This approach is **equivariant** to any 1:1 transformation of the attribute, say $T(a(\mathcal{P}))$.

- For an increasing function $T(\cdot)$ the corresponding interval for

$$[T(a_{lower}), T(a_{upper})]$$

- For an decreasing function $T(\cdot)$ the corresponding interval for

$$[T(a_{upper}), T(a_{lower})]$$

- So, we only need to determine the values a_{lower} and a_{upper} once for $a(\mathcal{P})$ and we have them for any $T(a(\mathcal{P}))$.

Example: Average Shark Length

```
sharks <- read.csv("../Data/sharks.csv")
popSharks <- rownames(sharks)

set.seed(341)
n <- 6
S <- getSample(1:65, n, replace = FALSE)
S
```

```
## [1] 9 20 14 63 13 59
```

```
Pstar <- S
B <- 10^4
set.seed(341)
Sstar <- sapply(1:B, FUN = function(b) getSample(Pstar, n, replace = T))

dim(Sstar)
```

```
## [1] 6 10000
```

```
Sstar[,1:10]
```

```
##      [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10]
## [1,]    9    9   14   20   63   63   13   63   13    63
## [2,]   63   13   14   14   63   13   20    9   13    63
## [3,]   59   20   13   20   63   63    9   20   20    63
## [4,]   59    9    9   20   13   14    9   14   14    59
## [5,]   13   63   59    9    9   20   14   63    9    59
## [6,]   14   14   20   14   63   20   14   14   63    63
```

```
avesBootSamp <- sapply(1:B, FUN = function(i) mean(sharks[Sstar[,i], "Length"]))
```

- Using the quantiles from the bootstrap distribution (based on 10,000 resamples from the sample) we obtain the following estimate of the interval that covers 95% of the bootstrap averages: (98.17, 179)
- Note, the population average is 151.86

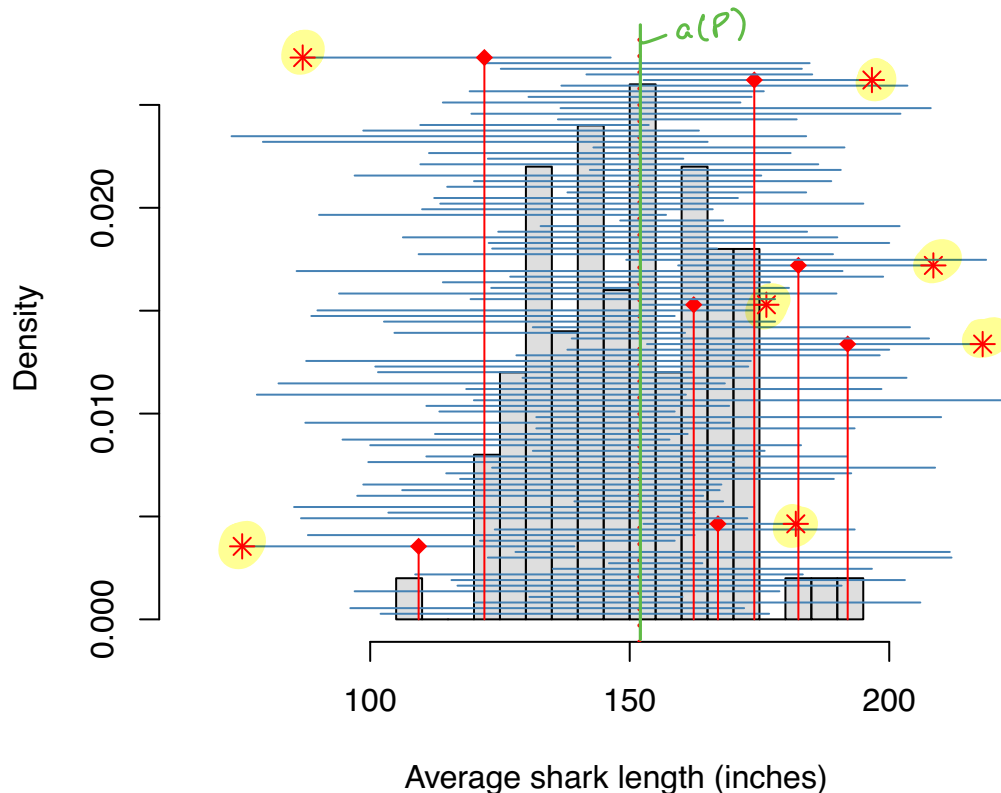
quantile(avesBootSamp, p = c(0.025, 0.975))

Estimating Coverage Probability: The Percentile Method

- To estimate the coverage probability associated with the percentile method (using a 95% confidence level) we
 - generate $B = 1000$ samples of size $n = 6$ and for each sample calculate $a_b = a(S_b^*)$.
 - and then from the values $a_1, \dots, a_B$, find $a_{lower} = Q_a(0.025)$ and $a_{upper} = Q_a(0.975)$
 - the 95% confidence interval for each sample is $[a_{lower}, a_{upper}]$.

- These 100 intervals (and their coverage) are illustrated below

100 individual 95% bootstrap confidence intervals



- As can be seen, 93 of the 100 bootstrap intervals actually cover the average in the population.

Comments:

- Simplicity is the attraction of this method, and explains its continued popularity.
- This method is transformation equivariant
- * The coverage probability is often incorrect if the distribution of the estimator is not nearly symmetric.
 - Coverage may be improved by considering “bias-corrected” versions of the percentile method. This, however, is outside the scope of this course. For more information on these alternatives see Section 11.3 in the book [Computer Age Statistical Inference](#) by Efron and Hastie.

and if the bootstrap sampling distribution is far from $a(P)$.

This can be improved by increasing both n and B . The more bootstrap samples the better the approximation of the tails of the sampling distribution and hence the better the performance of percentile method C.I's.

Summary

- We reviewed that the bootstrap distribution is an approximation to the sampling distribution for an attribute.
- We can use that approximation directly to construct a confidence interval.

Exercises

From the Inference Exercises try The Percentile interval part of problem 5.